Sophie Mirabai Woodward

PhD Candidate in Biostatistics
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Education

Harvard University
PhD in Biostatistics (In Progress) *2022–2027*
Developing methods to address unmeasured spatial confounding in causal inference under the guidance of Dr. Francesca Dominici.

Harvard College
BA in Statistics and Mathematics *2018–2022*
Graduated Magna Cum Laude with Highest Honors.

Publications

Woodward, S.M., Dominici, F., & Zubizarreta, J. (2025). Understanding Spatial Regression Models from a Weighting Perspective in an Observational Study of the Impact of Superfund Cleanups on Birth Outcomes. *Arxiv preprint*.

Woodward, S.M., Tec, M., & Dominici, F. (2024). An Instrumental Variables Framework to Unite Spatial Confounding Methods. *Under review*.

Tec, M., Trisovic, A., Audirac M., **Woodward, S.M.**, Hu, J., Khoshnevis, N. & Dominici, F. (2024). SpaCE: The Spatial Confounding Environment. *International Conference on Learning Representations*.

Woodward, S.M., Mork, D., Wu, X., Hou, Z., Braun, D., & Dominici, F. (2023). Combining aggregate and individual-level data to estimate individual-level associations between air pollution and COVID-19 mortality in the United States. *PLOS Global Public Health*, 3(8), e0002178.

Woodward, S.M., & Wu, X. (2023). Review 1: "Long-term Outdoor Air Pollution and COVID-19 Mortality in London: an Individual-level Analysis". *Rapid Reviews Infectious Diseases*.

Field, R. D., Moelis, N., Salzman, J., Bax, A., Ausiello, D., **Woodward, S.M.**, & Edwards, D. A. (2021). Inhaled water and salt suppress respiratory droplet generation and COVID-19 incidence and death on US coastlines. *Molecular Frontiers Journal*, 5(01n02), 17-29.

Awards

2025: Tom Ten Have Award at the *American Causal Inference Conference*.

2024: Certificate of Distinction in Teaching Award, Harvard Biostatistics.

2024: Selected student speaker at the *Accelerate: Data for Social Impact Conference*.

2022: National Science Foundation Graduate Research Fellowship.

Teaching

Statistical Inference I (BST 231) **Harvard University**
Teaching Fellow *2024–2025*
Supervisor: Dr. Rui Wang.

StatStart Summer Program **Harvard University**
Instructor *2024–2025*

Introduction to Probability and Statistics **Harvard University**
Teaching Assistant *2020–2022*
Supervisor: Dr. Joseph Blitzstein.

Skills

Programming: Python, R